

Surge Harness

Overcome Your Limits Safely

Elevator Pitch

Surge is a **wearable resistance system** designed to help climbers improve movement control safely by preventing hyperextension while strengthening stability muscles through doing what they love most, climbing.

The Team



Jordyn Rabinowitz
PM and Prototyping



Arva Syed
Biomechanics and User Testing

Why are we making this?

Jordyn : Rock Climber in Recovery

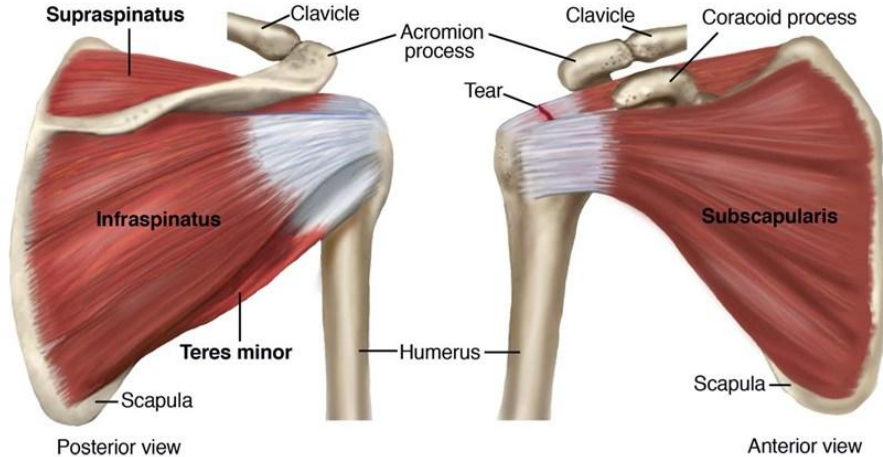
I found traditional weighted vests and PT bands to be restrictive or require boring repetitive exercises. This made it difficult to keep up with the necessary stability muscle exercises.

- **90%** of climbers report sustaining an upper extremity injury – fingers, followed by shoulder/arm
- **Female climbers** tend to be more hypermobile which leads them to being more at risk

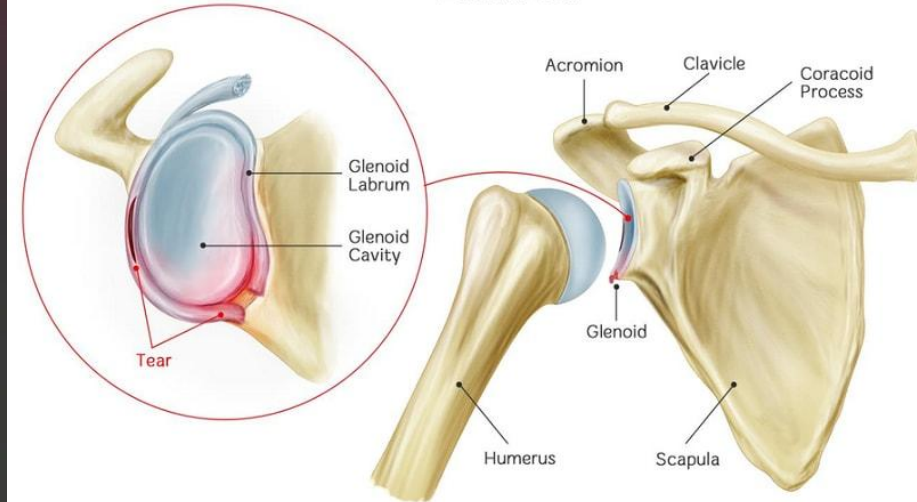
Source: nih.gov

Rotator Cuff Injury & Labrum Tears

Rotator Cuff Muscles



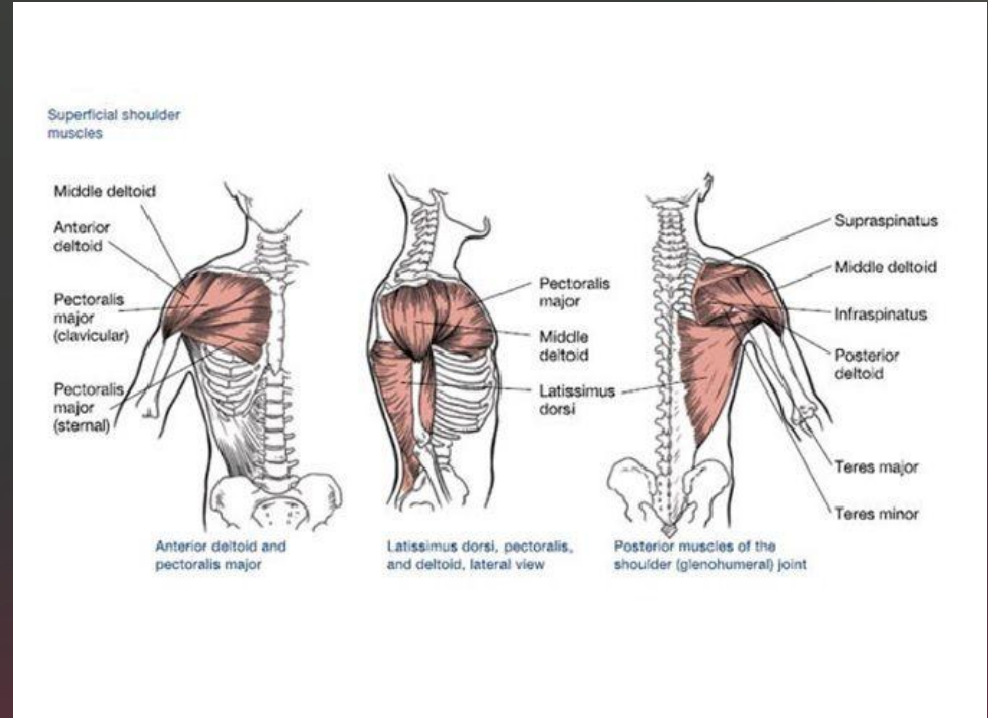
Labrum Tear



Solution

To prevent injury you need to improve strength, improve technique, and limit hyperextension.

Surge uses directional elastic bands anchored from the back to the lower bicep to slow movements, improve precision, and stabilize the shoulders, back, and core.

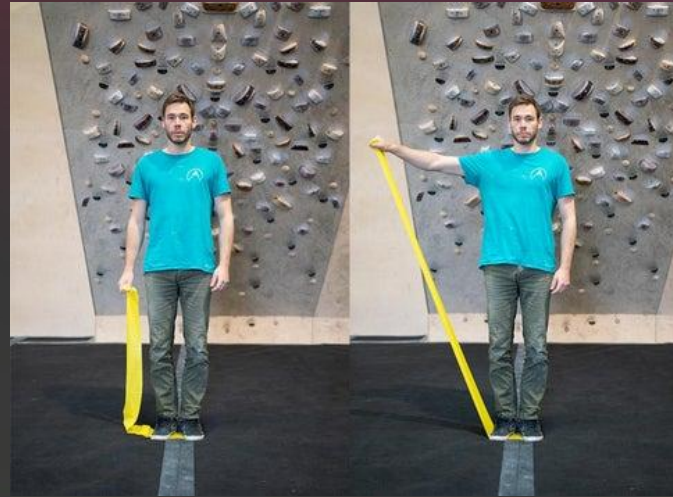
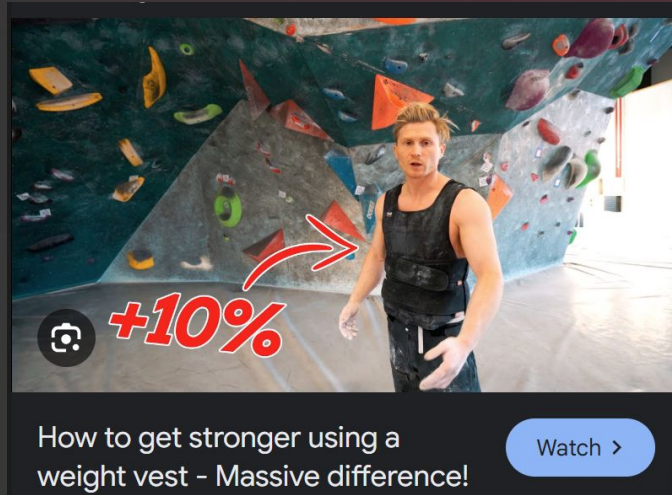


The Harness

- Modular limb wraps
- Bands anchored mid back→ shoulder → lower biceps
- Soft, breathable materials, adjustable straps



Climbing Focused Precedence



More Precedences

01

Stoko



02

The Body
Braid

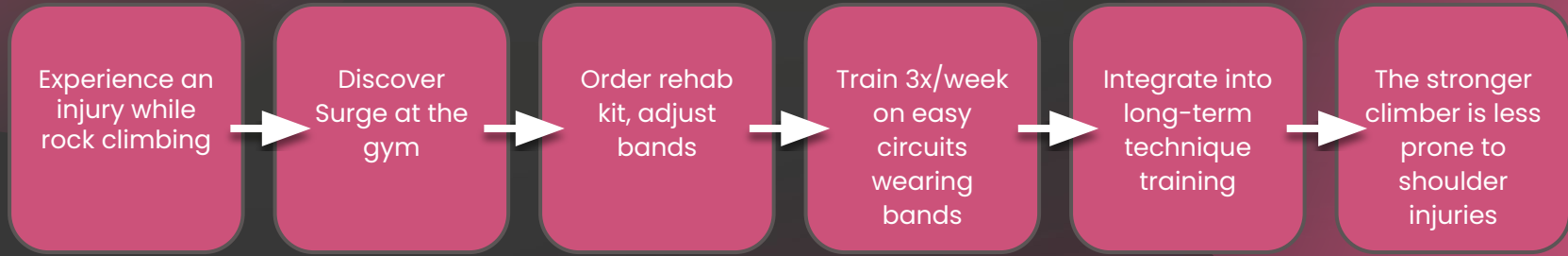


03

KT Tape

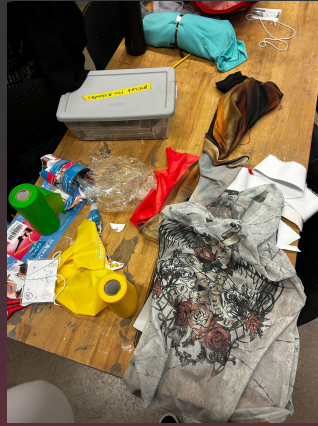


User Journey



Feasibility Testing

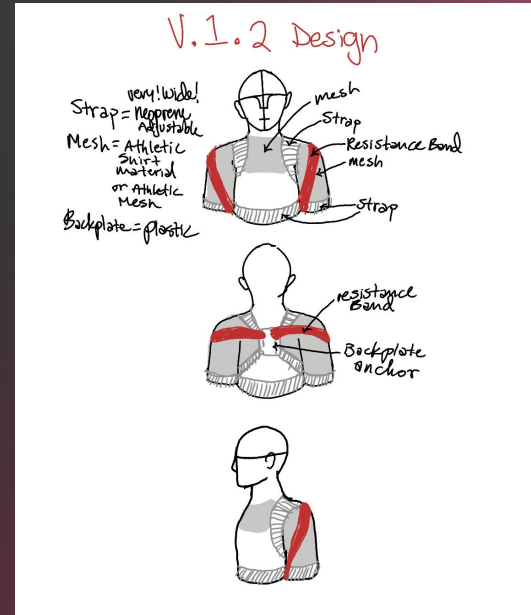
1. Testing different ways to engage muscles for rotator cuff stability using KT Tape
2. Testing different materials
3. Testing early prototype design by making it with thrifted materials fitted to the body and wearing it for 1 hr





What We Learned

1. **Fabric:** mesh and synthetic sports material
2. **Elastic:** Non-latex x-light resistance bands (RPE 2-3) with fabric overlay
3. **Hardware:** Low-profile plastic buckles
4. **Design:** Refine band routing to engage desired muscles and support the shoulder effectively



Implementation and User Testing

- **Weeks 1–2: Engage with the community**
- **Weeks 3–7: Focus group testing and professional review**
 - Test prototypes with focus group
 - Update prototype design
 - Receive feedback from PTs on new prototype
 - Repeat
 - Round 1 : Comfort/Durability
 - Round 2 : Performance during Rock Climbing
 - Round 3 : Daily Use
- **Weeks 8–9: Larger climbing event**
- **Week 10 : Final Deliverables**

Next Steps...

- **Apply for microgrants + reach out to rock climbing companies**
- **Have physical therapists review first prototype's anchor points**
- **Recruit research participants and professional reviewers**

Thank You!

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Implementation and User Testing

Weeks 1-2: Engage with the community

- Talk to climbers and physical therapists (PTs) in boulder climbing gyms about their experiences with shoulder injuries. Collect data via surveys.
- Recruit 5 climbers and two PTs for focus group testing
- Create updated prototypes for focus group testing

Implementation and User Testing

Weeks 3–7: Focus group testing and professional review

- 3 rounds
 - Test prototypes with focus group
 - Update prototype design
 - Receive feedback from PTs on new prototype
 - Repeat
- Round 1 : Comfort/Durability
- Round 2 : Performance during Rock Climbing
- Round 3 : Daily Use

Implementation and User Testing

Weeks 8–9: Larger climbing event

- Throw a rock climbing event at a local gym where people try out our polished wearable
- Participants will complete rock climbing challenges with and without our wearable
- Collect feedback to measure interest and collect performance metrics to see effectiveness of the wearable
- Run statistical analysis on collected data

Week 10 : Final Deliverables

- Present Polished wearable used at event and data visualizations showing how well (hopefully) it works

Problem vs solution

Statistical analysis was used to study association between participants' demographics and injuries. **Results:** A total of 397 participants responded to the survey. Mean age was 32.5 years with males comprising 85%. No significant differences in demographics or climbing behaviors were found between males and females. Ninety percent of participants reported sustaining an upper extremity injury. Fingers were the most common injury followed by shoulder/arm and elbow/forearm. Our study found females to be more likely to report a rock climbing-related injury, and more likely to undergo surgery for it. **Conclusion:** Female rock climbers were significantly more likely to report a shoulder/upper arm injury and were also more likely to report undergoing surgery compared with males, where these differences were not due to age or climbing behaviors. Further investigation is warranted into the association between shoulder injuries and female athletes to determine how the gender differences relate to extent of injury as well as health service utilization behaviors.

Abstract

Purpose: The purpose of this study was to identify gender-related differences in glenohumeral (GH) joint laxity, stiffness, and generalized joint hypermobility in healthy men and women.

Methods: Fifty-one healthy men and women were tested for generalized joint hypermobility, and anterior-posterior (AP) joint laxity and stiffness using a single-group factorial design.

Results: Women exhibited significantly more anterior joint laxity (men 8.3 ± 2.2 mm vs women 11.4 ± 2.8 mm, $P < 0.001$) and less anterior joint stiffness (men 20.5 ± 5.0 N x mm(-1) vs women 16.3 ± 4.2 N x mm(-1), $P < 0.01$) than men. Men had significantly more posterior joint laxity than anterior (Ant 8.3 ± 2.2 mm vs Post 9.6 ± 2.9 mm; $P < 0.001$), and women also had significantly less anterior joint stiffness than posterior (Ant 16.3 ± 4.2 N x mm(-1) vs Post 22.1 ± 6.9 N x mm(-1); $P < 0.01$). Women also demonstrated significantly more generalized joint hypermobility than men (men 1.0 ± 1.7 vs women 2.9 ± 2.1 ; $P = 0.02$).

Conclusions: Our findings may indicate a possible increased risk for instability in women, especially those participating in sports that require repetitive overhead-throwing motion. Future investigations should seek to determine the contribution of increased GH joint laxity and decreased joint stiffness to various injury states and examine these variables in other populations such as overhead-throwing athletes.

[PubMed Disclaimer](#)

Google search results for "how to prevent hyperextension of the shoulder for climbing".

AI Overview

To prevent shoulder hyperextension in climbing, **focus on engaging shoulder and scapular muscles, improving body positioning, using your lower body more, and strengthening your rotator cuff through proper warm-ups, strength training, and technique refinement.** Actively pull your shoulders down and away from your ears, and maintain good posture with your trunk upright and shoulder blades squeezed together.

On the wall

- **Engage your shoulder and scapular muscles:** Actively pull your shoulders down and away from your ears to keep the joint stable, rather than letting it hang passively.
- **Grip the hold firmly:** A tight grip will engage the rotator cuff muscles, helping to stabilize the joint.
- **Maintain good posture:** Keep your trunk upright and your shoulder blades gently squeezed together to create a strong foundation and reduce strain on your shoulders.
- **Shift your weight:** Position your body directly under your arm when possible to keep stress off the joint.

A Climber's Guide to Shoulder Pain: Treatment & Prevention ...
Jul 26, 2023 — Control your load-volume: Capacity is built over a long time. When you ar...
Mend Colorado

Navigating Shoulder Health in Rock Climbing Enthusiasts
Jan 18, 2024 — Proper Warm-Up: Prioritize a thorough warm-up routine that includes...
Wildhawk Physical Therapy

Climbing with Shoulder Hypermobility
Aug 14, 2025 — However, if you must engage in this motion, here are a few techniques to...
The Climbing Doctor

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Next Steps...

